

Draft - expert review pending

Compiled with problems:

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

ERROR in ./docs/semester-1/ggur-300/unit-03/topic-01.mdx

Module build failed (from ./node_modules/@docusaurus/mdx-loader/lib/index.js):

Error: MDX compilation failed for file

["/home/a2ahs/mega book for B.Ed/.claude/worktrees/agent-affcb48d7818e8183/docs/semester-1/ggur-300/unit-03/topic-01.mdx"](#)

Cause: Unexpected character `=` (U+003D) before name, expected a character that can start a name, such as a letter, `\$`, or `\_`

Details:

```
{
  "column": 36,
  "file": "",
  "message": "Unexpected character `=` (U+003D) before name,
    expected a character that can start a name, such as a
```

```
letter, `$`, or `_`",
```

```
"line": 141,
```

```
"name": "141:36",
```

```
"place": {
```

```
  "_bufferIndex": 33,
```

```
  "_index": 0,
```

```
  "line": 141,
```

```
  "column": 36,
```

```
  "offset": 7642
```

```
},
```

```
"reason": "Unexpected character `=` (U+003D) before name,
    expected a character that can start a name, such as a
    letter, `$`, or `_`",
```

```
"ruleId": "unexpected-character",
```

```
"source": "micromark-extension-mdx-jsx",
```

```
"url": "https://github.com/micromark/micromark-extension-
mdx-jsx#unexpected-character-at-expected-expect"
```

```
}
```

A four-step problem-solving strategy

Understand, plan, carry out, look back – and loop until the answer survives the real-world check.

1. Understand

what is asked,

2. Make a plan

choose a method

or a model

3. Carry out

compute carefully,

keep the units

4. Look back

check the answer

against the question

if the check fails, return to step 1

Mr. Danish is the timetable in-charge at a middle school in Larkana. The headteacher says: "Three teachers are on leave every Friday, and Classes 6, 7 and 8 still need their math periods. Fix it." Mr. Danish does not panic. He writes down what is asked (every class gets its math period on Friday) and what is given (6 math periods to cover, 3 teachers short). He tries a plan: combine two classes for a math activity so one teacher covers both. He carries it

out on paper, then looks back: Class 8's math period now clashes with the science lab slot. He returns to step one and adjusts. By the third loop, the timetable works. His method matters more than his arithmetic, and the method is what this topic teaches.

Explanation

Logical reasoning

The move	In one sentence	Everyday example
Spot the pattern	find what repeats or grows, and let it predict the next	each new matchstick triangle adds 2 sticks, so predict on
If-then reasoning	two stated facts force a conclusion that must hold	if the van seats 30 and 34 signed up, one trip is not enough
Deduction	apply a general rule to a particular case	every prime number has exactly two factors, so 17 has two
Elimination	rule out options until one remains	not the cupboard, not the shelf, not borrowed: the book is lost

textbook.com.pk

textbook.com.pk

Logical reasoning is thinking in ordered, checkable steps: each conclusion follows from what came before, so another person can retrace your path. The table above shows the four moves you will use most often. Two of them deserve a closer look.

If-then reasoning. "If the van seats 30, and 34 pupils signed up, then one trip cannot carry everyone." The conclusion is forced by the two facts. Quantitative reasoning leans heavily on this move, because numbers make the "if" precise. If the school's water tank holds 500 litres and the school uses about 100 litres a day, then a full tank lasts about five days. Notice that if-then reasoning only as

good as its starting facts: if the tank actually holds 400 litres, the five-day conclusion collapses even though the logic was perfect.

Elimination. "The missing book is not in the cupboard (checked), not on the shelf (checked), not borrowed by anyone (register says so); therefore it is lost." Elimination rules out options until one remains. It is how a teacher narrows down why a class's average fell: not the syllabus (same as last term), not absenteeism (attendance was normal), so most likely the test's hardest chapter fell in the exam week.

A common misconception to correct. Some trainees believe "there is one correct method for every problem". Real problems have many defensible routes. What makes a route correct is that each step follows from the last and the final answer survives the look-back check, not that it matches the textbook's method. Steen (2001) makes this point at the level of a whole society: the value of quantitative reasoning is not in ritual procedures but in sound judgement.

Problem-solving strategies

A problem-solving strategy is an ordered way through a problem you have not met before. The flowchart above shows the classic four steps, often traced to the mathematician George Polya: understand the problem, make a plan, carry out the plan, and look back.

Worked example. Problem: a school of 240 pupils must seat 4 pupils per bench in the hall for a function; how many benches are needed?

1. **Understand:** we need the number of benches; given 240 pupils and 4 per bench.
2. **Plan:** divide total pupils by pupils per bench.
3. **Carry out:** 240 divided by 4 is 60 benches.
4. **Look back:** does 60 make sense? 60 benches of 4 give exactly 240 seats. But the chief guest and two teachers also need seats, so 62 is safer. The look-back step is where the real-world answer separates from the bare arithmetic answer.

The four steps are also a teaching tool. When a pupil says "I don't know how to start", they are stuck at step one, not at the arithmetic. Asking "what is asked,

what is given?" is usually enough to unstick them, and it models the strategy in front of the whole class.

Activity: Spot the broken step

Work in pairs. Your tutor will show three short pieces of school reasoning, each with one flawed step (for example: "Attendance was 90 percent on Monday and 90 percent on Tuesday, so the same pupils were absent both days"). For each one, identify which logical move is being misused, name the broken step, and write the corrected version in two sentences. Then join another pair and compare repairs. About 20 minutes.

Check your understanding

1. Name the four steps of the problem-solving strategy in order, in your own words. (*Remember*)
2. A teacher reasons: "If every pupil brings 50 rupees, and 40 pupils promise to bring it, we will collect 2,000 rupees." State the two facts this if-then conclusion depends on, and one way the conclusion could fail. (*Apply*)
3. Use elimination in two sentences to narrow down why a class's water bill doubled this month, given: no new taps installed, same number of pupils, and the meter was read correctly. (*Analyze*)

Summary

Logical reasoning moves in checkable steps: spotting patterns, if-then chains, deduction, and elimination. Its strength depends on the truth of its starting facts. The four-step problem-solving strategy (understand, plan, carry out, look back) turns reasoning into a repeatable method, and the look-back step is where a bare answer becomes a real-world decision.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can name and use the four logical-reasoning moves with everyday examples.
- ☐ I can apply the four-step problem-solving strategy to a problem I have not seen before.
- ☐ I can find the broken step in a flawed piece of reasoning and repair it.

Try this at your practicum school

Take one real problem your cooperating teacher solved this week (a seating plan, a worksheet count, a duty roster). Reconstruct their solution as the four steps: understand, plan, carry out, look back. Note which step took them the longest, and ask them whether they agree with your reconstruction.

Summative task

A school notice says: "Last year 20 pupils failed the Class 8 math exam. This year we bought a new workbook series, and only 10 pupils failed. Therefore the new workbooks halved our failure rate." Write a short analysis that (a) names the reasoning move used, (b) identifies at least two other quantities that changed between the two years and could explain the drop, and (c) states what evidence would be needed before crediting the workbooks.

Mini-rubric (each criterion 0-2 marks):

- *Move named*: the analysis correctly identifies the if-then (cause-effect) structure.
- *Alternative explanations*: at least two real alternative quantities are named with a sentence each on how they could produce the same drop. **(Analyze)**
- *Evidence standard*: the analysis states what comparison or data would isolate the workbook effect.

Further reading

- Steen, L. A. (Ed.). (2001). *Mathematics and Democracy: The Case for Quantitative Literacy*. National Council on Education and the Disciplines.

<https://archive.org/details/mathematicsdemoc0000unse> (cited at bibliographic level; the scan is controlled-lending)

- Sikko, S. A. (2023). What can we learn from the different understandings of mathematical literacy? *Numeracy*, 16(1). <https://eric.ed.gov/?id=EJ1450768>

Tip: select any text on this page first to comment on that specific passage - or leave nothing selected for general feedback.

Give feedback